ECON 0100 | Vignette E2 | Competition

Due in Recitation. Vignettes are a certificate of the work done together in Recitation.

Q1 | A Higher Marginal Cost

The Witching Hour has a marginal cost of $MC = 10 + 2q$. Many identical workshops sell identical cauldrons, so each takes the market price as given.

a) At $P = 50$ what quantity maximizes the firm's profit?

b) Calculate the firm's maximal profit if $ATC = 50$ at the profit-maximizing quantity.

Now suppose the Ministry taxes every cauldron 10 coins, so marginal cost rises: $MC = 20 + 2q$. Start with identical firms earning zero economic profit in long-run equilibrium.

c) What happens to price, each firm's output, and profit after the new cost?

d) In the long run the price rises to 60. What quantity maximizes each remaining firm's profit?

e) What happens in the long run to price, each remaining firm's output, total market output, and the number of firms?